

Report Period	Sep 26, 2026 22:21 - Sep 26, 2026 22:27
Generated At	2026-09-26 22:27:37
Classification	ML: Isolation Forest + Random Forest

1. Executive Summary

		Traffic Summary		Count	
Metric	Value	Total Flows Observed		16321	
Total Threats Mitigated	14	Total Packets Analyzed		9930357346	
High Priority	7	Malicious Packets		9895631408	
Low Priority	7	True Negatives Passed		11263	
Manual Releases	0				
Manual Blocks	0	Attack Vector		Count	
		ICMP Flood		0	
ML Metrics	Value	SYN Flood		0	
False Positives	0	UDP Flood		11	
FP Rate	0.00%	Uncertain		3	
IF Precision	100.00%				
IF Recall	95.87%	Action		Count	
IF F1-Score	97.89%	Quarantined		3	
IF Accuracy	98.66%	Time Ban		3	
		Blackhole		0	
		Released		0	

2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	100.00%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	95.87%	Share of actual attacks successfully flagged
F1-Score	97.89%	Balanced measure combining Precision and Recall
Accuracy	98.66%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.00%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	4.13%	Actual attacks that went undetected
True Positive Rate (TPR)	95.87%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	100.00%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 5058	FN 218
Actual: Normal	FP 0	TN 11045

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	100.00%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	100.00%	Share of actual attacks successfully flagged
F1-Score	100.00%	Balanced measure combining Precision and Recall
Accuracy	100.00%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.00%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	0.00%	Actual attacks that went undetected
True Positive Rate (TPR)	100.00%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	100.00%	Normal traffic correctly identified as safe

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	0	0	0
Act: ICMP	0	0	0
Act: UDP	0	0	5058

2c. Hybrid Cascaded

Metric	Value	Description
Precision	100.00%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	95.87%	Share of actual attacks successfully flagged
F1-Score	97.89%	Balanced measure combining Precision and Recall
Accuracy	98.66%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.00%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	4.13%	Actual attacks that went undetected
True Positive Rate (TPR)	95.87%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	100.00%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 5058	FN 218
Actual: Normal	FP 0	TN 11045

2d. Response Latency

Metric	Value	Description
Detection Time	573.5 ms	Average time taken to detect an attack
Mitigation Response Time	1.2 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2e. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	1.09%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	3.61%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	3.56%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.20	4	3	2026-09-26 22:24:30	2026-09-26 22:27:10
10.0.0.21	4	3	2026-09-26 22:24:37	2026-09-26 22:27:11
10.0.0.22	4	3	2026-09-26 22:24:32	2026-09-26 22:27:11

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-26 22:24:30	10.0.0.20	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 22:24:32	10.0.0.22	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 22:24:37	10.0.0.21	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 22:24:41	10.0.0.20	DDoS	UDP Flood	80.8%	High	Time Ban (30s)
2026-09-26 22:24:41	10.0.0.20	DDoS	UDP Flood	80.8%	High	Time Ban
2026-09-26 22:24:43	10.0.0.22	DDoS	UDP Flood	85.0%	High	Time Ban (30s)
2026-09-26 22:24:43	10.0.0.22	DDoS	UDP Flood	85.0%	Critical	Time Ban
2026-09-26 22:24:48	10.0.0.21	DDoS	UDP Flood	78.5%	High	Time Ban (30s)
2026-09-26 22:24:48	10.0.0.21	DDoS	UDP Flood	83.8%	High	Time Ban
2026-09-26 22:25:41	10.0.0.20	DDoS	UDP Flood	80.8%	High	Time Ban (1m)
2026-09-26 22:25:43	10.0.0.22	DDoS	UDP Flood	85.0%	High	Time Ban (1m)
2026-09-26 22:25:48	10.0.0.21	DDoS	UDP Flood	83.8%	Critical	Time Ban (1m)
2026-09-26 22:27:11	10.0.0.22	DDoS	UDP Flood	86.0%	Critical	Time Ban (2m)
2026-09-26 22:27:11	10.0.0.21	DDoS	UDP Flood	87.0%	Critical	Time Ban (2m)

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.21	UDP Flood	0.6313	87.0%	Critical	Phase 2	1	10s	2026-09-26 22:27:11	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6362	86.0%	Critical	Phase 2	1	10s	2026-09-26 22:27:11	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6448	75.6%	Critical	Phase 2	1	10s	2026-09-26 22:27:10	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6454	83.8%	Critical	Phase 2	3	1m 11s	2026-09-26 22:25:48	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6135	85.0%	High	Phase 2	3	1m 11s	2026-09-26 22:25:43	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6426	80.8%	High	Phase 2	3	1m 11s	2026-09-26 22:25:41	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6456	83.8%	High	Phase 2	2	41s	2026-09-26 22:25:18	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6391	85.0%	High	Phase 2	2	41s	2026-09-26 22:25:13	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.20	UDP Flood	0.6452	80.8%	High	Phase 2	2	41s	2026-09-26 22:25:11	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6458	78.5%	High	Phase 2	1	11s	2026-09-26 22:24:48	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6391	85.0%	High	Phase 2	1	11s	2026-09-26 22:24:43	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6448	80.8%	High	Phase 2	1	11s	2026-09-26 22:24:41	Attack persisted - escalating to time ban

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE